

## 7 Differentiation

Continuity tells us that a function has no sudden “jumps,” but it does not describe the rate at which the function changes. This is the role of derivatives. They answer the question:

*How fast does  $f(x)$  change as  $x$  changes?*

Think of driving a car. Distance traveled is a function of time, while the speedometer shows how quickly the distance is changing at that exact moment. This is exactly what a derivative represents: the **rate of change at a given instant**.

In other words, a derivative tells us how much the output changes when the input changes by a tiny amount. Mathematically, this idea is expressed using a **limit**, which gives the precise definition of the derivative.

### 7.1 Differentiability of Functions

**Definition 7.1.** Let  $(a, b)$  be an interval and  $f : (a, b) \rightarrow \mathbb{R}$  be a function. Let  $c \in (a, b)$ . We say  $f$  is **differentiable at  $c$**  if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists. This limit is called the **derivative** of  $f$  at  $c$  and is denoted as  $f'(c)$ .

Corresponding to a function  $f$ , we define its derivative  $f'$  as the function whose domain consists of all points  $x$  at which the derivative of  $f$  exists.

**Example 7.1.** 1. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be defined by  $f(x) = 3x + e$ . Then  $f$  is differentiable at every point of  $\mathbb{R}$ . For any  $c \in \mathbb{R}$ , we compute

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} \frac{3x + e - (3c + e)}{x - c} = \lim_{x \rightarrow c} \frac{3(x - c)}{x - c} = 3.$$

Thus, the derivative exists and  $f'(c) = 3$ , for all  $c \in \mathbb{R}$ .

2. Let

$$f(x) = \begin{cases} x, & \text{if } x < 0, \\ x + 1, & \text{if } x \geq 0. \end{cases}$$

Here  $f$  is not differentiable at  $x = 0$ . First note  $f(0) = 0 + 1 = 1$ . Consider the difference quotient

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x) - 1}{x}, \quad x \neq 0.$$

Now for  $x > 0$ ,  $f(x) = x + 1$ , hence

$$\lim_{x \rightarrow 0^+} \frac{f(x) - 1}{x} = \lim_{x \rightarrow 0^+} \frac{x + 1 - 1}{x} = 1.$$

If  $x < 0$ ,  $f(x) = x$ , hence

$$\frac{f(x) - 1}{x} = \frac{x - 1}{x} = 1 - \frac{1}{x}.$$

As  $x \rightarrow 0^-$ , we have  $1/x \rightarrow -\infty$ , so

$$\lim_{x \rightarrow 0^-} \frac{f(x) - 1}{x} = +\infty,$$

hence, the left-hand limit diverges. Therefore, the limit does not exist, so  $f$  is not differentiable at  $x = 0$ .

**Theorem 7.2.** Let  $(a, b)$  be an open interval and a function  $f : (a, b) \rightarrow \mathbb{R}$  be differentiable at a point  $c \in (a, b)$ . Then  $f$  is continuous at  $c$ .

*Proof.* For all  $x \in (a, b)$ , but  $x \neq c$ ,

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c).$$

Since  $f$  is differentiable at  $c$ ,

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists and the limit is finite. Also, since  $\lim_{x \rightarrow c} (x - c) = 0$ , we have

$$\lim_{x \rightarrow c} [f(x) - f(c)] = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \cdot (x - c) = f'(c) \cdot 0 = 0,$$

since  $f'(c)$  is finite. Therefore

$$\lim_{x \rightarrow c} f(x) = f(c)$$

and this shows that  $f$  is continuous at  $c$ . □

**Remark 7.3.** The continuity of  $f$  at a point  $c \in (a, b)$  does not ensure differentiability of  $f$  at  $c$ . For example, let  $f(x) = |x|$ ,  $x \in \mathbb{R}$ . At  $x = 0$ ,  $f(x) = 0$ . Also  $f$  is continuous at 0.

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1.$$

Since the left-hand and right-hand derivatives of  $f$  are different, hence  $f$  is not differentiable at 0. Hence continuity at a point  $c$  does not imply differentiability at  $c$ .

**Example 7.4.** 1. Let

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Then  $f$  is differentiable at 0 and  $f'(0) = 0$ .

Indeed,

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \rightarrow 0} x \sin(1/x) = 0.$$

But

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), \quad x \neq 0.$$

Since  $\cos(1/x)$  oscillates as  $x \rightarrow 0$ , the derivative has no limit at 0. Hence  $f'$  is not continuous at 0.

2. Consider

$$f(x) = \begin{cases} x + x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$

Then  $f$  is differentiable on  $\mathbb{R}$  with

$$f'(0) = 1,$$

but

$$f'(x) = 1 + 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), \quad x \neq 0.$$

Again,  $f'$  oscillates near 0 and hence is not continuous there.

**Theorem 7.5 (Algebra of Differentiation).** Let  $f, g : (a, b) \rightarrow \mathbb{R}$  be differentiable at  $c \in (a, b)$ , and let  $\lambda \in \mathbb{R}$ . Then:

1. (**Linearity**)

$$(f + g)'(c) = f'(c) + g'(c), \quad (\lambda f)'(c) = \lambda f'(c).$$

2. (**Product rule**)

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

3. (**Quotient rule**) If  $g(c) \neq 0$ , then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

**Theorem 7.6 (Chain Rule).** Let  $f : (a, b) \rightarrow \mathbb{R}$  be differentiable at  $c \in (a, b)$ , and let  $g$  be defined on an interval containing  $f((a, b))$  and differentiable at  $f(c)$ . Then the composition  $g \circ f$  is differentiable at  $c$ , and

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

**Example 7.7.** 1. Let  $f(x) = x^2$  and  $g(y) = \sin y$ . Then

$$(g \circ f)(x) = \sin(x^2).$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \cos(x^2) \cdot (2x) = 2x \cos(x^2).$$

2. Let  $f(x) = e^x$  and  $g(y) = \ln y$ , defined on  $(0, \infty)$ . Then

$$(g \circ f)(x) = \ln(e^x) = x.$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \frac{1}{e^x} \cdot e^x = 1,$$

which matches the fact that the derivative of  $x$  is 1.

3. Let  $f(x) = \sqrt{x}$  on  $(0, \infty)$  and  $g(y) = y^3$ . Then

$$(g \circ f)(x) = (\sqrt{x})^3 = x^{3/2}.$$

By the Chain Rule,

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = 3(\sqrt{x})^2 \cdot \frac{1}{2\sqrt{x}} = 3x \cdot \frac{1}{2\sqrt{x}} = \frac{3}{2}\sqrt{x}.$$

## 7.2 Applications

Suppose a function starts and ends at the same height. Do you expect its slope to be zero at some point in between? The answer is affirmative. This remarkable result is known as Rolle's Theorem. The theorem is named after the French mathematician Michel Rolle, who proved the result for polynomials in 1691 without using differential calculus. A rigorous proof in the modern setting was later given by Augustin-Louis Cauchy in 1823.

**Theorem 7.8 (Rolle's Theorem).** *Let  $f : [a, b] \rightarrow \mathbb{R}$  be a function satisfying:*

1.  $f$  is continuous on  $[a, b]$
2.  $f$  is differentiable on  $(a, b)$
3.  $f(a) = f(b)$

*Then there exists  $c \in (a, b)$  such that  $f'(c) = 0$ .*

*Proof.* Since  $f$  is continuous on  $[a, b]$ ,  $f$  is bounded on  $[a, b]$ . Let

$$\sup_{x \in [a, b]} f(x) = M, \quad \inf_{x \in [a, b]} f(x) = m.$$

By the property of continuity there exists a point  $c$  in  $[a, b]$  such that  $f(c) = M$  and there exists point  $d$  in  $[a, b]$  such that  $f(d) = m$ . Two cases arise.

**Case 1.**  $M = m$ .

In this case  $f(x) = M$  for all  $x \in [a, b]$ . Therefore  $f'(x) = 0$  for all  $x \in [a, b]$ . The theorem holds trivially in this case.

**Case 2.**  $M \neq m$ .

In this case at least one of  $M$  and  $m$ , if not both, must be unequal to  $f(a)$  (and  $f(b)$ ). Let  $M \neq f(a)$ . Then  $c \neq a, c \neq b$ .  $\therefore a < c < b$ . By condition (ii),  $f'(c)$  exists. If possible, let  $f'(c) > 0$ . Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} > 0.$$

Hence there exists a positive  $\delta$  such that

$$\frac{f(x) - f(c)}{x - c} > 0$$

for all  $x \in [a, b]$  satisfying  $0 < |x - c| < \delta$ . This implies  $f(x) - f(c) > 0$  for all  $x \in [a, b]$  satisfying  $c < x < c + \delta$ . i.e.,  $f(x) > M$  for all  $x \in [a, b]$  satisfying  $c < x < c + \delta$ . This contradicts that  $M$  is the supremum of  $f$  on  $[a, b]$ . Consequently,  $f'(c) \neq 0 \dots$  (i) If possible, let  $f'(c) < 0$ . Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} < 0.$$

Hence there exists a positive  $\delta$  such that

$$\frac{f(x) - f(c)}{x - c} < 0$$

for all  $x \in [a, b]$  satisfying  $0 < |x - c| < \delta$ . This implies  $f(x) < f(c)$  for all  $x \in [a, b]$  satisfying  $c - \delta < x < c$ . i.e.,  $f(x) < M$  for all  $x \in [a, b]$  satisfying  $c - \delta < x < c$ . This contradicts that  $M$  is the supremum of  $f$  on  $[a, b]$ . Consequently,  $f'(c) \neq 0 \dots$  (ii)

From (i) and (ii) it follows that  $f'(c) = 0$ . Therefore  $\xi = c$ . If however,  $M = f(a) = f(b)$ , then  $m \neq f(a)$  and therefore  $d \neq a, d \neq b$  and  $a < d < b$ . Proceeding with similar arguments we can prove  $f'(d) = 0$ . Therefore  $\xi = d$ . This completes the proof of the theorem.  $\square$

**Remark 7.9.** 1. The conditions of Rolle's theorem are sufficient, but not necessary. Consider

$$f(x) = |x| + |x - 1|, \quad x \in [-1, 2].$$

Then

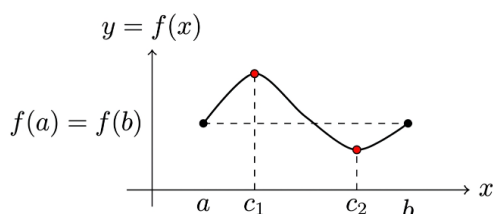
$$f(x) = \begin{cases} 1 - 2x, & -1 \leq x < 0, \\ 1, & 0 \leq x \leq 1, \\ 2x - 1, & 1 < x \leq 2. \end{cases}$$

The function  $f$  is continuous on  $[-1, 2]$ , but is not differentiable at  $x = 0, 1$ . Also,  $f(-1) = f(2) = 3$ . Thus,  $f$  does not satisfy all the conditions of Rolle's theorem. However,

$$f'(x) = 0, \quad 0 < x < 1.$$

Hence,  $f'(\xi) = 0$  for some  $\xi \in (-1, 2)$ , even though the hypotheses of Rolle's theorem are not all satisfied.

2. Geometrically, the theorem guarantees the existence of a point where the tangent to the curve  $y = f(x)$  is horizontal. Physically, if a particle returns to its initial position, then at some intermediate time its instantaneous velocity must be zero.



**Example 7.10.** 1. The hypotheses of Rolle's Theorem are sufficient but not necessary. That is, a function may satisfy the conclusion of the theorem even if one or more of its hypotheses fail. Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = x^3$ . On the interval  $[-1, 1]$ , we have  $f(-1) \neq f(1)$ , so the hypothesis  $f(a) = f(b)$  of Rolle's Theorem is not satisfied. Nevertheless,

$$f'(x) = 3x^2,$$

and hence  $f'(0) = 0$ . Thus, the conclusion of Rolle's Theorem holds even though one of its hypotheses fails.

2. Consider the equation  $x^3 + x + 1 = 0$ . Suppose  $a \neq b$  are two solutions. By Rolle's Theorem, there exists  $c \in (a, b)$  such that

$$3c^2 + 1 = 0,$$

which is impossible. Hence the equation has at most one real solution.

**Theorem 7.11 (Mean value theorem (Lagrange)).** Let a function  $f : [a, b] \rightarrow \mathbb{R}$  be such that (i)  $f$  is continuous on  $[a, b]$ , and (ii)  $f$  is differentiable at every point of  $(a, b)$ .

Then there exists at least a point  $\xi$  in  $(a, b)$  such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi).$$

*Proof.* Let us define  $\phi : [a, b] \rightarrow \mathbb{R}$  by

$$\phi(x) = f(x) + Ax, \quad x \in [a, b] \text{ where } A \text{ is a constant.}$$

Clearly,  $\phi$  is continuous on  $[a, b]$ , since  $f$  is continuous on  $[a, b]$ ; and  $\phi$  is differentiable at every point of  $(a, b)$ , since  $f$  is differentiable at every point of  $(a, b)$ . Let us choose  $A$  such that  $\phi(a) = \phi(b)$ . Then  $f(a) + Aa = f(b) + Ab$  and this determines

$$A = \frac{f(b) - f(a)}{a - b}.$$

For this choice of  $A$ ,  $\phi$  satisfies all conditions of Rolle's theorem on  $[a, b]$ . Therefore there exists at least a point  $\xi$  in  $(a, b)$  such that  $\phi'(\xi) = 0$ . But  $\phi'(\xi) = f'(\xi) + A$  and therefore

$$0 = f'(\xi) + \frac{f(b) - f(a)}{a - b},$$

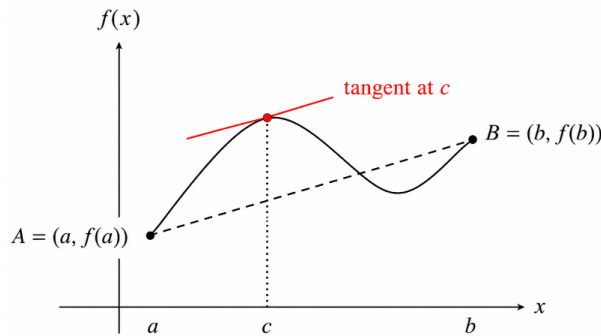
or,

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

□

**Remark 7.12.** Geometrically, it means that the tangent to the curve at  $x = c$  is parallel to the secant line joining  $(a, f(a))$  and  $(b, f(b))$ , that is,

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



**Example 7.13.** 1. Show that

$$|\sin x - \sin y| \leq |x - y|$$

for all  $x, y \in \mathbb{R}$ .

2. Show that for every  $x > 0$ ,

$$\ln(1 + x) < x.$$

3. Show that

$$|e^x - e^y| \leq e^{\max\{x, y\}} |x - y|$$

for all  $x, y \in \mathbb{R}$  has at most one real solution.

4. Suppose  $f : J \rightarrow \mathbb{R}$  is differentiable on an interval  $J$  and

$$f'(x) = 0 \quad \text{for all } x \in J.$$

Show that  $f$  is constant on  $J$ .

5. Let  $f : [2, 5] \rightarrow \mathbb{R}$  be continuous and differentiable on  $(2, 5)$ . Assume that

$$f'(x) = (f(x))^2 + \pi \quad \text{for all } x \in (2, 5).$$

True or false:  $f(5) - f(2) = 3$ .

6. Let  $f : (0, 1] \rightarrow \mathbb{R}$  be differentiable and suppose that

$$|f'(x)| < 1 \quad \text{for all } x \in (0, 1].$$

Define

$$a_n = f\left(\frac{1}{n}\right).$$

Show that the sequence  $(a_n)$  converges.

7. Suppose that  $f : J \rightarrow \mathbb{R}$  is differentiable on an interval  $J$  and satisfies

$$0 < m \leq f'(x) \leq M, \quad \forall x \in J,$$

where  $m, M > 0$ . Show that for every  $x, y \in J$  with  $x < y$ ,

$$m(y - x) \leq f(y) - f(x) \leq M(y - x).$$

### Solution.

1. Assume  $x \neq y$ . By the Mean Value Theorem, there exists  $c$  between  $x$  and  $y$  such that

$$\sin x - \sin y = \cos(c)(x - y).$$

Taking absolute values,

$$|\sin x - \sin y| = |\cos(c)||x - y| \leq |x - y|.$$

2. Apply the Mean Value Theorem to  $f(t) = \ln t$  on  $[1, 1+x]$ . Then there exists  $c \in (1, 1+x)$  such that

$$\ln(1+x) - \ln 1 = \frac{1}{c}x.$$

Since  $c > 1$ , we have  $\frac{1}{c} < 1$ . Hence

$$\ln(1+x) = \frac{x}{c} < x.$$

3. By the Mean Value Theorem, there exists  $c$  between  $x$  and  $y$  such that

$$e^x - e^y = e^c(x - y).$$

Since  $e^c \leq e^{\max\{x, y\}}$ , the result follows.

4. Let  $x, y \in J$  with  $x < y$ . Since  $J$  is an interval,  $[x, y] \subseteq J$ . By the Mean Value Theorem, there exists  $c \in (x, y)$  such that

$$f(y) - f(x) = f'(c)(y - x).$$

Since  $f'(c) = 0$ , it follows that

$$f(y) = f(x).$$

As  $x$  and  $y$  are arbitrary,  $f$  is constant on  $J$ .

5. The statement is false. By the Mean Value Theorem, there exists  $c \in (2, 5)$  such that

$$f(5) - f(2) = f'(c)(5 - 2) = 3f'(c).$$

Since

$$f'(c) = (f(c))^2 + \pi > \pi,$$

we obtain

$$f(5) - f(2) > 3\pi > 3.$$

Hence the given statement is false.

6. By the Mean Value Theorem, for each  $n \geq 1$ , there exists  $c_n \in \left(\frac{1}{n+1}, \frac{1}{n}\right)$  such that

$$a_{n+1} - a_n = f\left(\frac{1}{n+1}\right) - f\left(\frac{1}{n}\right) = f'(c_n)\left(\frac{1}{n+1} - \frac{1}{n}\right).$$

Hence,

$$|a_{n+1} - a_n| \leq \left|\frac{1}{n+1} - \frac{1}{n}\right| = \frac{1}{n(n+1)}.$$

Therefore, for  $m > n$ ,

$$|a_m - a_n| \leq \sum_{k=n}^{m-1} |a_{k+1} - a_k| \leq \sum_{k=n}^{m-1} \frac{1}{k(k+1)} = \frac{1}{n} - \frac{1}{m} \leq \frac{1}{n}.$$

Since  $\frac{1}{n} \rightarrow 0$ , the sequence  $(a_n)$  is Cauchy in  $\mathbb{R}$  and so it converges.

7. By the Mean Value Theorem, there exists  $c \in (x, y)$  such that

$$f(y) - f(x) = f'(c)(y - x).$$

Since

$$m \leq f'(c) \leq M$$

and

$$y - x > 0,$$

we obtain

$$m(y - x) \leq f(y) - f(x) \leq M(y - x).$$

**Theorem 7.14 (Constant Function Theorem).** Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . If

$$f'(x) = 0 \quad \text{for all } x \in (a, b),$$

then  $f$  is constant on  $[a, b]$ .

*Proof.* Let  $x_1, x_2 \in [a, b]$  and  $a \leq x_1 < x_2 \leq b$ . Then  $f$  is continuous on  $[x_1, x_2]$  and differentiable on  $(x_1, x_2)$ . By the Mean Value Theorem, there exists a point  $\xi \in (x_1, x_2)$  such that

$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

But  $f'(\xi) = 0$  by hypothesis. Therefore,

$$f(x_2) = f(x_1).$$

Since  $x_1$  and  $x_2$  are arbitrary points in  $[a, b]$ ,  $f$  is constant on  $[a, b]$ . □



**Corollary 7.15.** Let  $f : [a, b] \rightarrow \mathbb{R}$  and  $g : [a, b] \rightarrow \mathbb{R}$  be both continuous on  $[a, b]$  and differentiable on  $(a, b)$ . If

$$f'(x) = g'(x) \quad \text{for all } x \in (a, b),$$

then  $f = g + c$ , where  $c \in \mathbb{R}$  is a constant.

**Theorem 7.16 (Monotonicity Theorem).** Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$ . Then:

1. If  $f'(x) \geq 0$ , for all  $x \in (a, b)$ , then  $f$  is monotone increasing on  $[a, b]$ .
2. If  $f'(x) \leq 0$ , for all  $x \in (a, b)$ , then  $f$  is monotone decreasing on  $[a, b]$ .

*Proof.* 1. Let  $x_1, x_2 \in [a, b]$  and  $a \leq x_1 < x_2 \leq b$ . Then  $f$  is continuous on  $[x_1, x_2]$  and differentiable on  $(x_1, x_2)$ . By the Mean Value Theorem, there exists a point  $\xi \in (x_1, x_2)$  such that

$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Since  $f'(\xi) \geq 0$ , and  $x_2 - x_1 > 0$ , we obtain

$$f(x_2) - f(x_1) \geq 0.$$

i.e.,

$$f(x_2) \geq f(x_1).$$

As  $x_1$  and  $x_2$  are arbitrary in  $[a, b]$ ,  $f$  is a monotone increasing function on  $[a, b]$ .

2. The proof is similar and is left to the reader.

□

**Remark 7.17.** Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then

$$\begin{cases} f'(x) > 0 & \text{for all } x \in (a, b) & \implies & f \text{ is strictly increasing on } [a, b], \\ f'(x) < 0 & \text{for all } x \in (a, b) & \implies & f \text{ is strictly decreasing on } [a, b]. \end{cases}$$

**Example 7.18.** 1. Prove that  $\frac{2x}{\pi} < \sin x$ , for  $0 < x < \frac{\pi}{2}$ .

2. Prove that

$$\left(1 + \frac{1}{x}\right)^x > \left(1 + \frac{1}{y}\right)^y, \text{ if } x, y \in \mathbb{R} \text{ and } x > y > 0.$$

3. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfy

$$|f(x) - f(y)| \leq (x - y)^2, \text{ for all } x, y \in \mathbb{R}.$$

Prove that  $f$  is a constant function on  $\mathbb{R}$ .

### 7.3 Solution.

1. Let  $f(x) = \frac{\sin x}{x}$ ,  $0 < x < \frac{\pi}{2}$ . Then

$$f'(x) = \frac{x \cos x - \sin x}{x^2}.$$

Since  $x < \tan x$ ,  $0 < x < \frac{\pi}{2}$ , we have  $x \cos x - \sin x < 0$ . Hence  $f'(x) < 0$ , and therefore  $f$  is strictly decreasing on  $(0, \pi/2)$ . Thus

$$f(x) > f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}.$$

Therefore,  $\frac{2x}{\pi} < \sin x$ ,  $0 < x < \frac{\pi}{2}$ .

2. Let

$$f(x) = \left(1 + \frac{1}{x}\right)^x, \quad x > 0.$$

Then

$$f'(x) = \left(1 + \frac{1}{x}\right)^x \left[ \log \left(1 + \frac{1}{x}\right) - \frac{1}{1+x} \right].$$

Define

$$\phi(x) = \log(1+x) - \frac{x}{1+x}, \quad x \geq 0.$$

Then

$$\phi'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0 \quad (x > 0).$$

Hence  $\phi$  is strictly increasing on  $[0, \infty)$ . Since

$$\phi(0) = 0,$$

we obtain

$$\log(1+x) > \frac{x}{1+x}, \quad x > 0.$$

Replacing  $x$  by  $1/x$  gives

$$\log \left(1 + \frac{1}{x}\right) > \frac{1}{1+x}, \quad x > 0.$$

Therefore,

$$f'(x) > 0, \quad x > 0.$$

Thus  $f$  is strictly increasing on  $(0, \infty)$ . Hence, if  $x > y > 0$ ,

$$\boxed{\left(1 + \frac{1}{x}\right)^x > \left(1 + \frac{1}{y}\right)^y}.$$

3. By the given condition,

$$-(x-y)^2 \leq f(x) - f(y) \leq (x-y)^2.$$

Fix  $c \in \mathbb{R}$  and put  $x = c + h$  and  $y = c$ . Then

$$-h^2 \leq f(c+h) - f(c) \leq h^2.$$

If  $h \neq 0$ , division by  $h$  gives

$$-h \leq \frac{f(c+h) - f(c)}{h} \leq h \quad (h > 0),$$

and

$$h \leq \frac{f(c+h) - f(c)}{h} \leq -h \quad (h < 0).$$

In either case, by the Squeeze Theorem,

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = 0.$$

Hence

$$f'(c) = 0.$$

Since  $c \in \mathbb{R}$  was arbitrary,

$$f'(x) = 0 \quad \text{for all } x \in \mathbb{R}.$$

By the Constant Function Theorem,  $f$  is constant on  $\mathbb{R}$ .

**Theorem 7.19 (Cauchy's Mean Value Theorem).** *Let  $f$  and  $g$  be continuous real-valued functions on  $[a, b]$  which are differentiable in  $(a, b)$ . Then there is a point  $x \in (a, b)$  at which*

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

*Proof.* Left as an exercise. □

**Theorem 7.20 (Darboux).** *Let  $I = [a, b]$  and a function  $f : I \rightarrow \mathbb{R}$  be differentiable on  $I$ . Let  $f'(a) \neq f'(b)$ . If  $k$  be a real number lying between  $f'(a)$  and  $f'(b)$  then there exists a point  $c$  in  $(a, b)$  such that  $f'(c) = k$ .*

**Example 7.21.** 1. Let  $f : [-1, 1] \rightarrow \mathbb{R}$  be defined by

$$f(x) = \begin{cases} 0, & x \in [-1, 0] \\ 1, & x \in (0, 1]. \end{cases}$$

Does there exist a function  $g$  such that

$$g'(x) = f(x), \quad x \in [-1, 1]?$$

If possible, let there exist a function  $g : [-1, 1] \rightarrow \mathbb{R}$  such that

$$g'(x) = f(x), \quad x \in [-1, 1].$$

Then  $g$  is differentiable on  $[-1, 1]$  and

$$g'(x) = \begin{cases} 0, & x \in [-1, 0] \\ 1, & x \in (0, 1]. \end{cases}$$

Since  $g$  is differentiable on  $[-1, 1]$  and  $g'(-1) \neq g'(1)$ , by Darboux's theorem  $g'$  must assume every real number lying between  $g'(-1)$  and  $g'(1)$ , i.e., between 0 and 1 on  $[-1, 1]$ . But this is not so and therefore  $g$  does not exist.

**Theorem 7.22.** *If  $f : [a, b] \rightarrow \mathbb{R}$  be differentiable on  $[a, b]$  then the derived function  $f'$  cannot have a jump discontinuity on  $[a, b]$ .*

## 7.4 Local Extrema

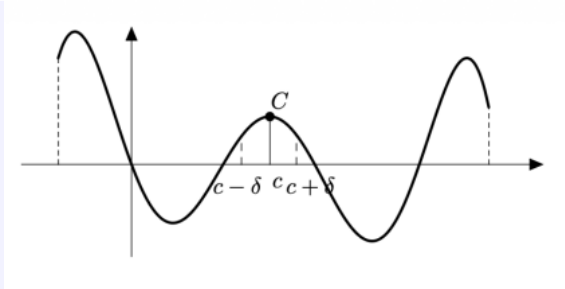
In the real world, systems fluctuate between peaks and valleys rather than moving in straight lines. Studying local maxima and minima allows us to locate these critical turning points where a system reaches an optimal high or low state. In business, they help maximize profit and minimize cost under realistic constraints. In machine learning, algorithms navigate error landscapes to find local minima and improve prediction accuracy. In physics, objects naturally settle at local energy minima for structural stability. Mathematically, these points mark where the rate of change is zero ( $f'(c) = 0$ ), turning abstract curves into practical, optimized decisions.

**Definition 7.2 (Local Maximum and Local Minimum).** *Let  $J \subseteq \mathbb{R}$  be an interval and let  $f : J \rightarrow \mathbb{R}$  be a function. A point  $c \in J$  is called a **local maximum** of  $f$  if there exists  $\delta > 0$  such that*

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \leq f(c) \quad \text{for all } x \in (c - \delta, c + \delta).$$



Similarly,  $c \in J$  is called a **local minimum** of  $f$  if there exists  $\delta > 0$  such that

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \geq f(c) \quad \text{for all } x \in (c - \delta, c + \delta).$$

A point  $c$  is called a **local extremum** if it is either a local maximum or a local minimum.

**Definition 7.3** (Global Maximum and Global Minimum). Let  $J \subseteq \mathbb{R}$  be an interval and let  $f : J \rightarrow \mathbb{R}$  be a function.

A point  $x_0 \in J$  is called a **global maximum** of  $f$  on  $J$  if

$$f(x) \leq f(x_0) \quad \text{for all } x \in J.$$

Similarly,  $x_0 \in J$  is called a **global minimum** of  $f$  on  $J$  if

$$f(x) \geq f(x_0) \quad \text{for all } x \in J.$$

Do you believe that there is any relation between a global maximum and a local maximum? For example, is any global maximum necessarily a local maximum? Or is any local maximum necessarily a global maximum?

**Example 7.23.** Consider the following functions on  $\mathbb{R}$ :

1. **Local maximum.** Let  $f(x) = -x^2$ . At  $c = 0$ , we have  $f(0) = 0$ . For  $x$  close to 0,  $f(x) = -x^2 \leq 0 = f(0)$ . Hence,  $c = 0$  is a local maximum point.
2. **Local minimum.** Let  $f(x) = x^2$ . At  $c = 0$ , we have  $f(0) = 0$ . For  $x$  close to 0,  $f(x) = x^2 \geq 0 = f(0)$ . Hence,  $c = 0$  is a local minimum point.
3. **Saddle point.** Let  $f(x) = x^3$ . At  $c = 0$ , we have  $f(0) = 0$ . But in every neighborhood of 0, there are points  $x > 0$  where  $f(x) > 0 = f(0)$  and points  $x < 0$  where  $f(x) < 0 = f(0)$ . Also  $f'(c) = 0$ . So  $c = 0$  is neither a local maximum nor a local minimum; it is a saddle point.

**Example 7.24.** Consider the function  $f : [a, b] \rightarrow \mathbb{R}$  defined by  $f(x) = x$ . Since  $f(x) \leq b = f(b)$  for every  $x \in [a, b]$ , the point  $b$  is a global maximum of  $f$ . However,  $b$  is not a local maximum, as it is an endpoint of the interval. Similarly,  $a$  is a global minimum of  $f$ , but it is not a local minimum.

**Theorem 7.25** (Fermat's Theorem). Let  $J \subseteq \mathbb{R}$  be an interval and let  $f : J \rightarrow \mathbb{R}$  be differentiable at  $c \in J$ . If  $c$  is a local extremum of  $f$ , then

$$f'(c) = 0.$$

*Proof.* Assume that  $c$  is a local maximum of  $f$ . Then, for  $h$  sufficiently small,

$$f(c+h) \leq f(c).$$

Hence,

$$\frac{f(c+h) - f(c)}{h} \leq 0 \quad \text{if } h > 0,$$

and

$$\frac{f(c+h) - f(c)}{h} \geq 0 \quad \text{if } h < 0.$$

Passing to the limit as  $h \rightarrow 0^+$  and  $h \rightarrow 0^-$ , we obtain

$$f'(c) \leq 0 \quad \text{and} \quad f'(c) \geq 0.$$

Therefore,

$$f'(c) = 0.$$

The proof for a local minimum is analogous. □

**Example 7.26.** The assumption that  $c$  is an interior point is essential in Fermat's Theorem. If  $c$  lies on the boundary of the domain, the conclusion  $f'(c) = 0$  may fail. For instance, let

$$f : [0, \infty) \rightarrow \mathbb{R}, \quad f(x) = x.$$

At  $c = 0$ , we have  $f(0) = 0$ , and clearly

$$f(x) \geq f(0) \quad \text{for all } x \in [0, \infty),$$

so  $f$  has a local minimum at 0. However,  $c = 0$  is not an interior point of the domain. Moreover,  $f'(0) = 1 \neq 0$ , which shows that the conclusion of Fermat's Theorem does not hold in this case.

**Theorem 7.27 (First derivative test for extrema).** Let  $f$  be continuous on  $I = [a, b]$  and  $c$  be an interior point of  $I$ . Let  $f$  be differentiable on  $(a, c)$  and  $(c, b)$ .

1. If there exists a neighbourhood  $(c - \delta, c + \delta) \subset I$  such that

$$f'(x) \geq 0 \quad \text{for } x \in (c - \delta, c), \quad f'(x) \leq 0 \quad \text{for } x \in (c, c + \delta),$$

then  $f$  has a local maximum at  $c$ .

2. If there exists a neighbourhood  $(c - \delta, c + \delta) \subset I$  such that

$$f'(x) \leq 0 \quad \text{for } x \in (c - \delta, c), \quad f'(x) \geq 0 \quad \text{for } x \in (c, c + \delta),$$

then  $f$  has a local minimum at  $c$ .

3. If  $f'(x)$  keeps the same sign on  $(c - \delta, c)$  and  $(c, c + \delta)$ , then  $f$  has no extremum at  $c$ .

*Proof.* 1. Let  $x \in (c - \delta, c)$ . Applying the Mean Value Theorem to  $f$  on  $[x, c]$ , there exists  $\xi \in (x, c)$  such that

$$f(c) - f(x) = (c - x)f'(\xi).$$

Since  $f'(\xi) \geq 0$ , we have

$$f(x) \leq f(c), \quad x \in (c - \delta, c).$$

Let  $x \in (c, c + \delta)$ . Applying the Mean Value Theorem to  $f$  on  $[c, x]$ , there exists  $\eta \in (c, x)$  such that

$$f(x) - f(c) = (x - c)f'(\eta).$$